CHALLENGE 2: DAY 31-60

LEARNING & FULL-STACK PROJECT ROADMAP

Target: Job-Ready MERN Developer

CURRENT STATUS	PRIMARY GOAL	CAPSTONE PROJECT	METHODOLOGY
Day 31 Completed	JS, React & MERN	DevCareer OS	Learn → Build → Test

Important Note: *DevTrack (Day 30)* remains your completed frontend project baseline. The next 30 days strengthen core full-stack foundations to construct a separate, backend-powered capstone project. Do not overwrite or break DevTrack.

PHASE 1 Days 31–37: JavaScript & React Foundations

DAY	TOPIC	LEARNING & PRACTICE FOCUS	OUTPUT STATE
Day 31	**Array & Object Mastery**	`map`, `filter`, `find`, `findIndex`, `some`, `every`, spread operator, destructuring	Completed
Day 32	**Reduce & Summaries**	Totals, averages, counting, grouping; calculate application statistics	Next Priority
Day 33	**Objects, Sets & Maps**	Deduplication, frequency maps, keyed lookup patterns	Practice
Day 34	**Strings & Array Solving**	Normalization, searching algorithms, sorting, handling edge cases	Practice
Day 35	**Async JavaScript**	Promises, `async/await` syntax, `try/catch` blocks, `Promise.all`	Practice
Day 36	**Browser & API Basics**	`fetch`, JSON serialization, HTTP status codes, request/response, errors	Practice
Day 37	**React State & Rendering**	Props vs State, rendering lifecycle, immutable updates, component boundaries	Mini Review

PHASE 2 Days 38–44: React, APIs & Authentication Concepts

DAY	TOPIC	LEARNING & PRACTICE FOCUS	OUTPUT STATE
Day 38	**Data Fetching in React**	Handling loading, success, error, and empty states; avoiding stale UI	Build Feature
Day 39	**Search, Filter & Sort**	Controlled inputs, derived state calculations, combining multi-filters	Build Feature
Day 40	**Pagination & UI Design**	Page state management, page boundary logic, reusable list/table components	Build Feature
Day 41	**Custom React Hooks**	`useEffect` dependencies, cleanup functions, building custom data hooks	Build Feature
Day 42	**Forms & Validation**	Controlled forms, inline validation messages, submit state handling	Build Feature
Day 43	**Authentication Concepts**	Login flows, JWT tokens, protected routes, auth state, client security	Design & Plan
Day 44	**API Mini-Project Review**	Combine API fetching, search/filter, error handling into a clean UI	Checkpoint

PHASE 3 Days 45–51: Node.js, Express & MongoDB Backend

DAY	TOPIC	LEARNING & PRACTICE FOCUS	OUTPUT STATE
Day 45	**Node.js & npm Basics**	Modules System, Environment Variables (`.env`), Scripts, Package Mgmt	Backend Setup
Day 46	**Express REST API**	Routing, Controllers, Middleware, Request/Response cycles, HTTP Codes	CRUD API
Day 47	**MongoDB & Mongoose**	Schema Design, Models, Data Validation, CRUD Database Operations	Database Setup
Day 48	**Validation & Error Handling**	Input Sanitization, Centralized Error Handling, Standardized API Responses	Harden API
Day 49	**JWT Authentication**	Password Hashing (bcrypt), Register/Login flows, Token verification	Auth Endpoints
Day 50	**Authorization & Security**	Role and ownership verification, Route Protection, Environment Secrets	Secure Routes
Day 51	**API Testing & Integration**	Postman Automated Tests, Edge-case coverage, Frontend Connection	Checkpoint

PHASE 4 Days 52–60: Full-Stack Capstone Project & Deployment

DAY	TOPIC	LEARNING & PRACTICE FOCUS	OUTPUT STATE
Day 52	**DevCareer OS Architecture**	Define Requirements, User Flows, Data Models, Wireframes, Task Breakdown	Planning
Day 53	**Backend Architecture**	Express Server, MongoDB setup, Core Data Schemas, Base CRUD APIs	Backend
Day 54	**Authentication System**	Auth Controllers, JWT Generation/Validation, Secure Route Guards	Backend
Day 55	**Frontend Foundation**	React Router Setup, Global Layout, Auth Context, Responsive Shell	Frontend
Day 56	**Career Management Features**	Job Tracking, Status Pipelines, Search/Filter, Interview Logs	Core Features
Day 57	**Learning & Analytics Dashboard**	Goal Tracker, Progress Summaries, Visual Metric Cards, Summary Stats	Core Features
Day 58	**UI Polish & Testing**	Validation Messages, Loading/Error/Empty States, Mobile/Accessibility Check	QA Audit
Day 59	**Deployment & Documentation**	Host Frontend & Backend, Configure Environment Vars, Draft Production README	Release
Day 60	**Final Assessment & Interview Prep**	System Architecture Walkthrough, Live Demo Prep, MERN Technical Q&A	Final Review

Capstone Project Blueprint: DevCareer OS

Core Features (MVP)

Essential

- **Authentication:** Secure Register/Login, JWT token-based session persistence.
- **Job Tracker:** Full CRUD for Applications (Company, Role, Status, Salary, Notes).
- **Interview Module:** Track rounds, dates, preparation notes, and feedback.
- **Learning Tracker:** Goals, completion status, skill metrics, and activity logs.
- **Dashboard:** Application totals, conversion stats, and upcoming schedules.

Technology Stack

MERN

- **Frontend:** React, React Router, Tailwind CSS / Custom CSS, Axios.
- **Backend:** Node.js, Express.js REST API Architecture.
- **Database:** MongoDB & Mongoose ORM for data schemas.
- **Security:** JSON Web Tokens (JWT), bcrypt password hashing.
- **Tools:** Postman API Testing, Git/GitHub, Cloud Deployment.

Daily Execution Strategy

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6	STEP 7
Learn	Practice	Build	Test	Explain	Document	Commit

Final Interview Readiness Checklist

- [ ] **JavaScript Fundamentals:** Mastered Array methods, Closures, Async/Await, Event Loop, Object immutability, and ES6+ features.
- [ ] **React & Client State:** Ability to explain State/Props, Hook lifecycles, Render optimizations, Controlled forms, and Routing mechanics.
- [ ] **Node & Express REST Architecture:** Clear explanation of Middleware chains, HTTP Status codes, Error handling, and REST standards.
- [ ] **MongoDB Schemas & Security:** Knowledge of Data Modeling, Mongoose queries, Password Hashing (bcrypt), and JWT Authorization headers.
- [ ] **Full-Stack Project Demo:** Deliver a concise 3-minute pitch detailing Problem statement, Stack choices, Architecture, and Key challenges overcome.